

Yuhan Chi

School of Mathematical Sciences, Fudan University · chi-shan0707.github.io · github.com/Chi-Shan0707

Profile

- Undergraduate at Fudan University (School of Mathematical Sciences), pursuing a double degree in Information and Computing Science and Artificial Intelligence.
- GPA: 3.96/4.00 (most recent semester); ranked 10th (top 5%) in cohort.
- Research interests: mathematical interpretability for AI trust, verification/evaluation of LLM reasoning, and reinforcement learning for sequential decision-making.

Education

- Fudan University, Shanghai, China
 - B.S. Candidate in Information and Computing Science & Artificial Intelligence (Double Degree), School of Mathematical Sciences
 - Sep. 2025 – Present
 - GPA: 3.96/4.00 (most recent semester)
 - Rank: 10th (top 5%) in cohort

Research Interests

- Mathematical Interpretability and AI Trust: grounding claims about model reliability in inspectable mathematical and statistical evidence
- Reasoning Verification and Evaluation: studying when surface-level signals from LLM traces are diagnostic, misleading, or protocol-dependent
- Reinforcement Learning and Sequential Decision-Making: using RL concepts to reason about uncertainty, feedback, and interaction with environments

Selected Open-Source Projects

- token-verification-mirage (<https://github.com/Chi-Shan0707/token-verification-mirage>) — AI4Math Workshop (<https://ai4math2026.github.io/>)
 - Solo-author project; led the full pipeline from dataset selection and model deployment to trace generation, evaluation design, analysis, figures, related-work organization, and writing
 - Poster, ICML 2026 Workshop on AI for Math (AI4Math)
 - Audits how evaluation protocol choices such as pooling, in-sample scoring, and direction-agnostic AUROC can change apparent verification performance
 - Shows that shallow token statistics can be useful diagnostics, but should not be treated as stable standalone verifiers without stronger controls
- code-not-text (<https://github.com/chi-shan0707/code-not-text>) — Interactive Demo (<https://chi-shan0707.github.io/code-not-text/demo>)
 - Solo project; cross-domain measurement study of cheap CoT-surface and token-trajectory features on DeepSeek-R1-0528-Qwen3-8B
 - Reports strong math signal (AoA 0.958, AUROC@100% 0.982), partial science signal (AoA 0.799), and weak coding transfer (AoA 0.434)
 - Tests coding-side robustness through an 83-scalar feature sweep, grouped

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- ablations, a CoT-only judge, nonlinear MLPs, SSL pre-training, and token-level de-knotting
- Frames the result as measurement non-invariance: the same feature family tracks convergence-like behavior in math but not executable correctness in coding
- TinyLoRA-GRPO-Coder (<https://github.com/Chi-Shan0707/TinyLoRA-GRPO-Coder>)
- Solo project; independent reimplementations and adaptations inspired by *Learning to Reason in 13 Parameters*
- Moved a small-parameter adaptation and GRPO-style training pipeline toward verifiable competitive-programming code generation on Qwen2.5-Coder-3B
- Uses compile-and-run rewards rather than static heuristics
- Built as a full research-system exercise: data processing, training, multi-GPU setup, reward design, evaluation, and validation
- microgpt.cpp (<https://github.com/Chi-Shan0707/microgpt.cpp>)
- Solo project; minimal GPT implementation from first principles in C++, built to understand model internals without relying on high-level frameworks

Academic Service

- Reviewer, ICML 2026 Workshop on AI for Math (AI4Math) (<https://ai4math2026.github.io/>), 2026

Skills

- Programming: Python, C++, C, Java, Node.js, Lean 4
- ML/AI: PyTorch, ML experimentation, LLM evaluation, reinforcement-learning basics
- Tools: Git, GitHub, Linux, LaTeX, Markdown, VS Code
- Other: Mathematical proof writing, technical writing, self-directed learning, competitive programming

Additional Information

- Maintains [github-unflag-playbook-cn](https://chi-shan0707.github.io/github-unflag-playbook-cn) (<https://chi-shan0707.github.io/github-unflag-playbook-cn>), a Chinese playbook documenting GitHub account flagging, recovery cases, and appeal workflows
- Contributor to [FDUGuideBook/nav-site](https://github.com/FDUGuideBook/nav-site) (<https://github.com/FDUGuideBook/nav-site>), a Fudan community site
- Personal website: chi-shan0707.github.io (<https://chi-shan0707.github.io>)
- GitHub: [Chi-Shan0707](https://github.com/Chi-Shan0707) (<https://github.com/Chi-Shan0707>)
- Actively self-studying mathematics, algorithms, and AI beyond formal coursework, with ongoing work documented in public repositories